Investigating Cross-Situational Word Learning with Similarity-Guided Contrastive Loss

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1 Introduction

The fundamental challenge of resolving referential ambiguity in word learning can be addressed by cross-situational word learning (CSWL). Cross-situational word learning (CSWL), the process of tracking word-referent co-occurrences across ambiguous situations, has been found in children aged 12-14 months (Smith and Yu, 2007) and adults (Yu and Smith, 2007). A CSWL model by Vong and Lake (2022) employs contrastive learning, aligning representations by pulling co-occurring words and referents together while pushing them apart from pairs of different trials. However, a key limitation of this approach is its strict application of contrastive learning. The model applies a uniform 'pushing' force to all non-matching scenes, operating on the implicit assumption that all contexts are equally dissimilar and should be separated to the same degree. Furthermore, by using dot-product similarity on the unnormalized representations of a pretrained visual encoder (VGG-16), the model may conflate semantic alignment with incidental magnitude biases. This approach is cognitively implausible since (1) sampling a memory with an object which is also present in the current scene results in pushing apart the words and its referents' embedding in their respective embedding spaces, (2) objects can be similar (e.g., craft-scissors and hospital scissors) but only appear together rarely. Applying a uniform loss on such items would push their representations apart, making the model blind to their visual similarity over time. A more cognitively plausible model might modulate the strength of the 'pushing' force based on the visual similarity between scenes. This would allow semantically or visually related contexts to maintain representational overlap, while still separating truly distinct situations.

This study tests whether modifying the contrastive loss to be sensitive to visual similarity and correcting for potential magnitude biases improves a model's ability to match the referential ambiguity effect and the Zipfian distribution advantage during learning.

How does modulating the contrastive learning with visual- similarity based scaling affect a model's ability to learn word-referent mappings compared to the unaltered baseline model as described by Vong and Lake (2022)?

RQ1: *Do the technical modifications (L2 normalization and SGCL) cause the model to exhibit performance patterns different from the human-like pattern demonstrated by the VL-Baseline model?*

H1: *The VL-Norm and SGCL-Norm models will show a referential ambiguity effect, with accuracy decreasing with increasing referential ambiguity.*

RQ2: *Do the technical modifications (L2 normalization and SGCL) alter the model to capture the human-like performance when learning under Zipfian distribution?*

H2:*The VL-Norm and SGCL-Norm models will show better performance on word-referent pairs learned under Zipfian distribution compared to those learned under uniform distribution with similar presentation frequency. The performance will increase with increasing presentation frequency.*

2 Methods

2.1 Data

A pool of 55 RGB images was compiled from the Office-31, Home-Office, and Fruits-360 datasets.

To provide the necessary variation in visual similarity for testing the SGCL's scaling mechanism, items were selected from several categories (e.g., food, kitchenware, office items, furniture) alongside random household objects. Selected images had clean, white backgrounds similar to those used by Vong and Lake (2022); any artificially inserted shadows were removed using a background removal tool. To match the format of the NOUN database (Vong and Lake, 2022), images were processed as follows: a binary mask was created using adaptive thresholding and morphological operations to isolate the object. The object was then centered, and scaled to ensure a white margin around the object resulting in a 224x224 image (see Appendix A.1).

2.2 Modelling Framework

The model from Vong and Lake (2022), is used as a baseline model. Its architecture consists of two encoders and a memory module. The image encoder uses a frozen VGG-16 model pretrained on ImageNet. The logits of the VGG-16 are passed to a trainable projection head that outputs 64-dimensional embedding vectors. The text encoder is a simple, randomly initialized embedding layer that maps novel word labels to 64-dimensional vectors. This study utilizes only the scene caption network by Vong and Lake (2022), where entire scenes are compared instead of breaking them down into individual word-referent combinations. During this process, three similarity matrices are created: (1) between all current referents and all current words (match), (2) between all current referents and all memory words (mismatch), and (3) between all memory referents and all current words (mismatch). The matrices are transformed into embeddings, which are used for the loss calculation.

The second model is based on Vong and Lake (2022) but adds L2- normalization to both encoders to mitigate magnitude effects during similarity assessment. This was done because both logit and confidence estimates of CNNs are highly sensitive to the properties of the training data. In settings with imbalanced class distributions, the normalized weights of the last convolutional layer strongly correlate with class frequency (Menon et al., 2020). These weights are used to calculate the logits, which serve as the direct input for generating confidence scores via the softmax function Even in relatively balanced datasets such as ImageNet (Deng et al., 2009),the VGG-16 training's use of cross entropy loss (Simonyan and Zisserman, 2014) makes it susceptible to confidence inflation (Lin et al., 2017). Even without overinflated confidence and balanced datasets, the models representational magnitude will still be shaped by the characteristics of the training dataset. A model trained on ImageNet showed a smooth, monotonically increasing relationship between predicted confidence and empirical accuracy (Xia et al., 2021). This is concerning since the data used has different characteristics than the VGG-16 training data, thus incidental magnitude patterns might skew dot-product-based alignment measures.

The third model addresses the cognitive implausibility of uniformly separating all non-matching scenes by introducing Similarity-Guided Contrastive Loss (SGCL). This loss scales the original loss's 'pushing' force based on visual similarity between scenes, implementing the idea that semantically related contexts should maintain representational overlap.

The loss scaling factor α is defined using softmax formulation to create the bounded scalar between 0 and 1. Thus, α can be described as a contextual distinctiveness factor that uses the current scenes internal similarity as reference.

$$\alpha = \frac{\exp(S_{\text{intra}}/\tau)}{\exp(S_{\text{intra}}/\tau) + \exp(S_{\text{cross}}/\tau)}$$

This simplifies to a sigmoid function of the difference between the similarity within the scene and the similarity across scenes:

$$\alpha = \sigma\left(\frac{S_{\text{intra}} - S_{\text{cross}}}{\tau}\right)$$

Therefore, separation of the contrastive loss is only reduced if cross-scene similarity exceeds the scene's internal similarity indicating that the two scenes are meaningfully similar.

The S_{intra} similarity is defined as the average cosine similarity of the image representations within the current scene under the exclusion of

self-similarity. S_{cross} was initially intended to be defined as the maximum cosine similarity of visual representations found between the scenes. However, the small number of objects within conditions caused frequent sampling of the same images in both the current and the memory scene. Therefore, S_{cross} is defined as the mean cosine similarity between all image embeddings in the current scene and all image embeddings in the memory scene. Thus, α approaches one (applying the full contrastive loss) when scenes are distinct and approaches zero (attenuating the loss) when scenes are highly similar. All equations are provided in Appendix A.2.

2.3 Modeling Procedure

To generate the scenes used for training and testing, images and their corresponding names were randomly selected from the data pool. As in Vong and Lake (2022), the names were randomly assigned across the selection. Unlike Vong and Lake (2022), trials were pre-generated rather than created before each run. This ensures that word-referent pairs and their presentation frequencies are identical across models. Preliminary tests showed larger performance variations when testing combinations of cross-dataset word-referent assignments (see Appendix A.3). Pre-generation controls for similar effects that could arise from different combinations within the same dataset.

For every condition, 20 independent runs were pre-generated using random selection of referents and corresponding words. In the referential ambiguity simulations, training on each run included 18 word-referent pairs that were presented six times each across scenes. Referential ambiguity was controlled by varying the number of 2–4 pairs per scene. In the Zipfian/uniform simulations, each run included 28 pairs with four pairs presented per scene. The pair presentation frequency followed either Zipfian (1–62 presentations) or uniform (10 presentations) distribution. During training, scenes were contrasted with previously seen scenes to compute contrastive loss. In the referential ambiguity evaluation three false referents were presented with the target referent. In the Zipfian evaluation, 27 false referents were presented next to the target. The selection consisted of forced-choice testing, where models chose the referent with maximum dot-product similarity between word and referent embeddings.

2.4 Statistical Analysis

ANOVAs were used to test the research questions. All analyses employed Type II sum of squares, with effect sizes quantified by partial eta squared (η_p^2). Assumption checks for normality (Shapiro-Wilk test) and homogeneity of variance (Levene's test) were conducted for each model. Where significant main effects or interactions were found, follow-up one-way ANOVAs and Tukey HSD tests were performed.

Referential Ambiguity (RQ1). A 3 (Model: VL-Baseline, VL-Norm, SGCL-Norm) × 3 (Referent Count: 2, 3, 4) between-subjects ANOVA tested effects on accuracy. Follow-up one-way ANOVAs and Tukey HSD tests examined the effect of referential ambiguity within each model and compared models at each level of ambiguity.

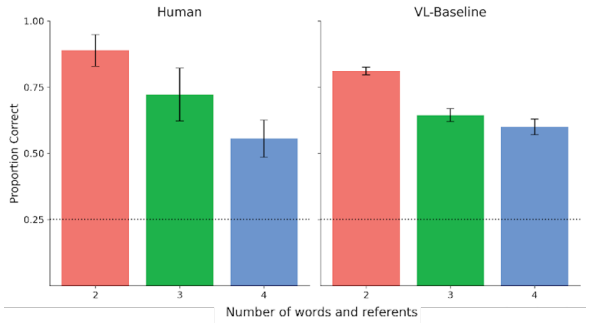
Zipfian Distribution (RQ2). Two ANOVAs were conducted. First, a 3 (Model) × 3 (Frequency Group: Low (1–5), Matched (8–11), High (19–62)) between-subjects ANOVA on accuracy within the Zipfian condition tested sensitivity to presentation frequency. Second, a 3 (Model) × 2 (Distribution: Zipfian vs. Uniform) between-subjects ANOVA on matched-frequency items tested for a Zipfian advantage. The same follow-up procedures (one-way ANOVAs, Tukey HSD tests) were applied to both analyses.

3 Results

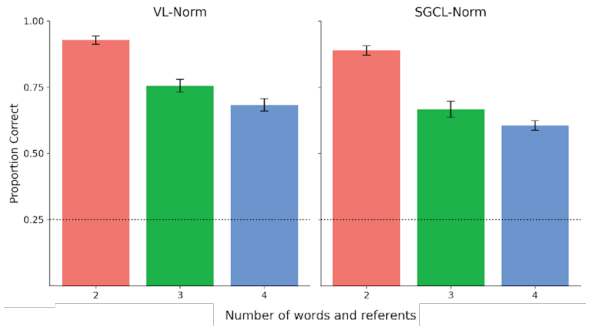
3.1 Referential Ambiguity

The human performance benchmarks extracted from Yu and Smith (2007) show that with increasing referential ambiguity the learning performance decreases (see Appendix A.4). The mean correct proportion was 88.9% with two referents, 72.2% with three and 52.6% with four. All three models qualitatively replicated this pattern (see Figure 1).

A 3×3 between-subjects ANOVA revealed significant main effects of both model type, $F(2, 171) = 16.46$, $p < .001$, $\eta p^2 = .161$, and number of referents, $F(2, 171) = 97.79$, $p < .001$, $\eta p^2 = .534$, on test accuracy. The interaction between model type and referent count



(a) Human vs. VL-Baseline



(b) VL-Norm vs. SGCL-Norm

Figure 1: Model performance under referential ambiguity. (Top) Comparison of human learning patterns with the baseline model. (Bottom) Performance comparison between normalized models.

was not significant, F(4, 171) = 0.80, p = .526, ηp^2 = .018. This indicates that models showed similar performance patterns in response to referential ambiguity (see Appendix A.4). Assumption checks indicated minor violations of normality (Shapiro-Wilk p = .043) and homogeneity of variance (Levene's p = .028); ANOVAs were considered robust to these deviations. One-way ANOVAs confirmed significant ambiguity effects for each model: VL-Baseline: $F(2, 57) = 22.34$, $p < .001$, $\eta_p^2 = .439$; VL-Norm: $F(2, 57) = 34.84$, $p < .001$, $\eta_p^2 = .550$; SGCL-Norm: $F(2, 57) = 43.27$, $p < .001$, $\eta_p^2 = .603$. Tukey HSD tests showed that for all models, 2-referent accuracy (VL-Baseline: M=.811; VL-Norm: M=.928; SGCL-Norm: M=.889) was significantly higher than both three-referent (all $p < .001$) and four-referent accuracy (all $p < .001$), but three- vs four-referent differences were not significant (all $p > .05$). It is unclear whether the human performance benchmarks showed significant pairwise differences. Thus, H1 is considered supported: all three models, exhibited the referential ambiguity effect.

Between-model one-way ANOVA showed that performances differed most in conditions with two word-referent pairs per scene, $F(2, 57) = 14.02$, $p < .001$, $\eta_p^2 = .330$. Overall, VL-Norm achieved the highest overall accuracy (M = 0.789, SD = 0.134), followed by SGCL-Norm (M = 0.720, SD = 0.149) and VL-Baseline (M = 0.685, SD = 0.136) (see Appendix A.3 for detailed results).

3.2 Zipfian Distribution Learning

Human benchmarks show two key patterns (via Vong and Lake, 2022; Hendrickson and Perfors, 2019): a frequency gradient within the Zipfian condition (Low: 26% < Matched: 53% < High: 77%) and a Zipfian advantage for matched-frequency items over the uniform condition (53% vs. 34.3%).

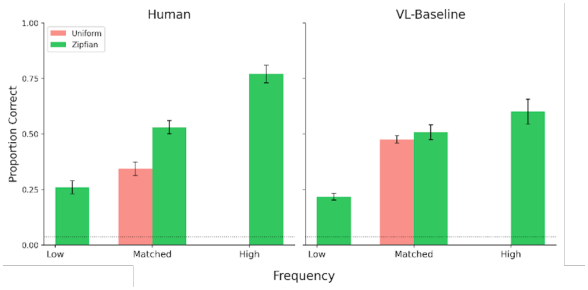
A 3 (Model) × 3 (Frequency: Low, Matched, High) ANOVA on Zipfian-condition accuracy revealed a significant interaction, $F(4, 171) = 2.84$, $p = .026$, $\eta_p^2 = .062$, indicating models responded differently to frequency (see Appendix A.5). Follow-up one-way ANOVAs within each model showed only the increase from Low to Matched frequency was significant (all $p < .001$); the increase from Matched to High was not (all $p > .05$).

Visually, VL-Baseline (Figure 2, top) most closely resembled the human pattern, though it lacked a significant high-frequency benefit. The normalized models (VL-Norm, SGCL-Norm) showed a weaker performances on high-frequency items (Figure 2, bottom). Models differed significantly only at the low frequency level, $F(2, 57) = 8.570$, $p < .001$, $\eta_p^2 = .231$.

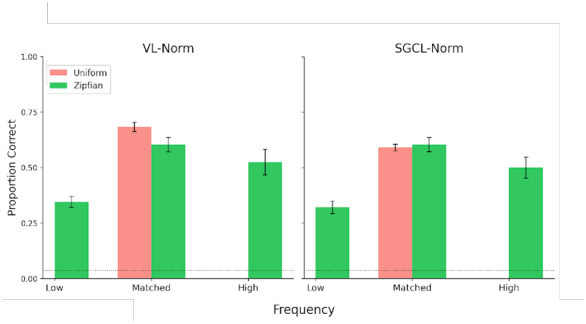
A 2 (Distribution: Zipfian vs. Uniform) × 3 (Model) ANOVA on matched-frequency items found no significant Zipfian advantage (main effect of Distribution: $F(1, 114) = 0.27$, $p = .605$, $\eta_p^2 = .002$; interaction: $F(2, 114) = 2.65$, $p = .075$, $\eta_p^2 = .044$). The only significant simple effect was that VL-Norm performed worse under Zipfian ($M = 0.604$, $SD = 0.145$) than Uniform ($M = 0.684$, $SD = 0.091$) (see Appendix A.5).

These results indicate that H2 was not supported. Contrary to the hypothesis, the VL-Norm and SGCL-Norm models did not show a significant Zipfian advantage, nor did they

show a significant performance increase from matched to high presentation frequencies. Only VL-Baseline showed a qualitative trend, but only the low-to-matched frequency increase was significant.



(a) Human vs. VL-Baseline



(b) VL-Norm vs. SGCL-Norm

Figure 2: Model performance under Zipfian distribution. (Top) Comparison of human learning patterns with the baseline model. (Bottom) Performance comparison between normalized models.

4 Discussion

This study posed two research questions. RQ1 asked if L2 normalization and similarity-scaling (SGCL) would alter human-like performance patterns. Results showed that, for the referential ambiguity effect, they did not (H1 supported). RQ2 asked if these modifications would improve the model's ability to capture human Zipfian learning. They did not (H2 not supported). The key finding is that L2 normalization (VL-Norm) dramatically improved overall accuracy, suggesting the baseline model's performance was significantly impaired by magnitude biases in the frozen VGG-16 encoder. This indicates that Vong and Lake (2022) model may have relied on architecturally convenient but cognitively implausible signals. While all models replicated the human referential ambiguity effect, only the unnormalized baseline showed a qualitative resemblance to human Zipfian learning patterns, and this advantage disappeared with normalization.

This suggests the baseline's ability to approximate some human patterns was not due to a human-like learning strategy, but an artifact of its representational space. The model likely exploited magnitude biases from ImageNet training, where logit magnitude correlates with class frequency and confidence. The interaction between magnitude and angular alignment during learning requires further research. It is possible that models strongly modulate representation magnitude to minimize contrastive loss. In such a representational space, representations within the same encoder might show close angular proximity but weak semantic relatedness. The application of normalization might destabilize the training due to more changes in the angular alignment of the representational spaces.

Contrastive loss, especially normalised, may be unsuitable for learning from Zipfian distributions because of the constant repelling of high-frequency items from varied contrastive scenes. Thus, stable semantic relationships would need to form despite the increased likelihood of appearing together with unrelated objects.

The failure of SGCL to improve upon simple normalization indicates that visual similarity is not an adequate proxy for the semantic relation. However, the performance might have been additionally deflated due to the lack of inherent structure of the randomly generated scenes.

This study has several limitations. The use of a fixed, classification-pretrained encoder (VGG-16) introduces unknown constraints. The dataset is small and the randomly generated scenes lack the semantic structure of real-world contexts. Future models should be tested on larger datasets and more naturalistic scenes. In addition, attempting to learned semantic similarity metrics for loss scaling might provide some insights into the potential magnitude of this learning signal.

4.1 Conclusion

Three models contrastive loss models: VL-Baseline, VL-Norm (with L2 normalization), and SGCL-Norm (with similarity-scaled loss) were tested on cross-situational word learning tasks. VL-Norm achieved the highest overall accuracy, significantly outperforming the baseline.

All models replicated the human referential ambiguity effect, with accuracy decreasing as the number of referents per trial increased. In the Zipfian learning condition, only VL-Baseline showed patterns similar to the human high frequency advantage and the advantage over uniform presentation, but the found differences were not statistically significant. Normalization eliminated Zipfian patterns while improving overall performance, suggesting that the resemblance to human performance is dependent on magnitude bias. These results are an indication, that the model's ability to match human behavioral patterns might not align with the common conceptualization that angular characteristics encode relational semantic meaning.

References

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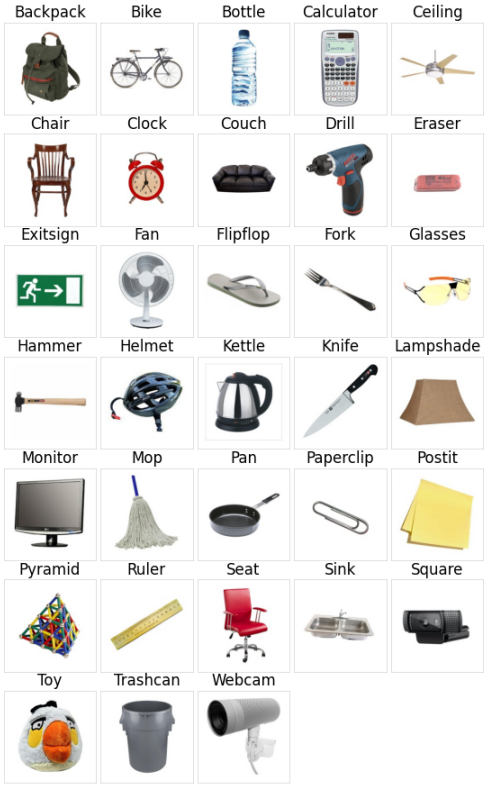
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A Appendix

A.1 Appendix One

(a) HomeOffice images (33 items)



(b) Fruits-360 images (11 items)



(c) Office-31 images (11 items)

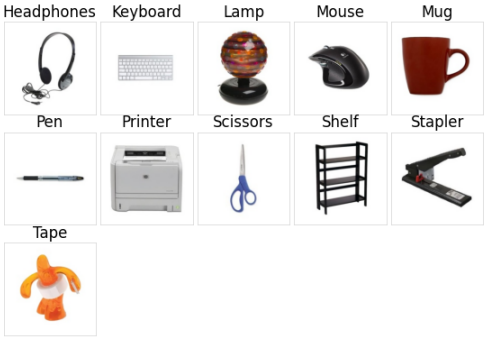


Figure 3: Complete set of visual stimuli used in experiments, organized by source dataset.

A.2 Appendix Two

Similarity Grounded Contrastive Loss

The loss function is defined as:

$$L = \max\big(0, s(v\prime, u) - s(v, u) + \eta\big) + \max\big(0, s(v, u\prime) - s(v, u) + \eta\big)$$

where the similarity function is:

$$s(V, U) = \frac{1}{M} \sum_{j=1}^{M} \max_{1 \le i \le N} (v_i \cdot u_j)$$

Variables:

- $V = \{v_1, \ldots, v_N\}$: visual input of the current scene
- $V\prime = \{v\prime_1, \ldots, v\prime_N\}$: visual input of the memory scene
- $U = \{u_1, \ldots, u_M\}$: linguistic input of current scene
- $U\prime = \{u\prime_1, \ldots, u\prime_M\}$: linguistic input of memory scene

Representation Normalization

L2 normalization is applied to normalize representation magnitude:

$$v_i = f_\theta(x_i) \quad \text{for } i = 1, \ldots, N$$

$$u_j = f_\phi(w_j) \quad \text{for } j = 1, \ldots, M$$

where $v_i, u_j \in \mathbb{R}^d$ with $\|v_i\| = \|u_j\| = 1$.

Input Data:

- $X = \{x_1, x_2, \ldots, x_N\}$: Visual Input
- $W = \{w_1, w_2, \ldots, w_M\}$: Linguistic Input

Implication for Similarity

Cosine similarity reduces to a dot product with normalization:

$$\text{sim}(v, u) = \frac{v \cdot u}{\|v\|\|u\|} = v \cdot u$$

Adaptation Loss Function

$$L_{\text{SGCL}} = \alpha \cdot \Big[\max\big(0, s(v\prime, u) - s(v, u) + \eta\big) + \max\big(0, s(v, u\prime) - s(v, u) + \eta\big)\Big]$$

where α is a scaling factor based on visual similarity within and between scenes.

Scaling Factor α:

$$\alpha = \frac{\exp(S_{\text{intra}}/\tau)}{\exp(S_{\text{intra}}/\tau) + \exp(S_{\text{cross}}/\tau)}$$

Intra-Scene Similarity:

$$S_{\text{intra}} = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} v_i \cdot v_j$$

Cross-Scene Similarity:

$$S_{\text{cross}} = \frac{1}{MN} \sum_{i=1}^{M} \sum_{j=1}^{N} v_i \cdot v_{j'}$$

A.3 Appendix Three

A baseline referential ambiguity trial was conducted to estimate the impact and feasibility of using common real world objects from this studies' datapool (POOL). The objects in the NOUN database (Horst and Hout, 2015) used by Vong and Lake (2022) were found to be perceived as distinct by participants. Since this study uses images of common objects, the representations of the VGG-16 will most likely encode more specific visual and semantic relationships. This might skew the performance of the VL-Model compared to the results during the implementation in Vong and Lake (2022). Therefore, cross-dataset word-referent assignments will be tested to determine whether real referents and words enable the model to cheat. To create similar conditions for the comparisons, the most visually distinct images were selected from the two datasets by using farthest point sampling on the representations of the final layer of the frozen VGG-16 encoder. The selected NOUN image subset showed an average cosine similarity of 0.271 (range: 0.091-0.431). The images selected from this study reached an average cosine similarity of 0.198 (range: 0.068-0.363). The referential ambiguity trials were conducted for the VL-Baseline and VL-Norm model under the scene caption implementation using three pairs per scene, six occurrences per pair and three foils + one target during the evaluation. In addition, the same 18 images were used across all 20 runs. Following cross-dataset word referent assignments were tested: Pool Names x Pool Images, Pool Names x VL Images, Pool Images x VL Names, and VL Images x VL Names. The words and referents were randomly paired and 20 runs were conducted to assess the performance. The results can be seen in the table below.

Table 1: Performance of VL-Baseline across different cross-dataset word-referent assignments

Condition		Mean	Min	Max	SD
Images	Names				
Pool	Pool	0.661	0.500	0.889	0.104
VL	Pool	0.747	0.556	0.889	0.075
Pool	VL	0.628	0.444	0.833	0.105
VL	VL	0.700	0.500	0.889	0.099

Using the VL-Baseline the tests indicate that real world objects are harder to learn. Trials that use Pool Images have the lowest mean accuracy. This might be because the real world objects are more visually distinct from one another or because the VGG-16 representation magnitudes are higher. The real world names, however, seem to improve the performance, as the trials that used Pool Names have higher performance when the same images were used.

Table 2: Performance of VL-Norm across different cross-dataset word-referent assignments

Condition		Mean	Min	Max	SD
Images	Names				
Pool	Pool	0.783	0.500	0.944	0.112
VL	Pool	0.831	0.722	1.000	0.081
Pool	VL	0.744	0.444	1.000	0.130
VL	VL	0.742	0.444	1.000	0.165

When using VL-Norm (normalised encoder representations), the performance improves across all trials. This indicates that the VL-Baseline model was affected by magnitude effects. The performance decreases observed when using real-world objects disappear with normalization. Pool names still improve performance across images. The performance improvements due to real-world names appear to be larger than those caused by the images. The performance gap observed between `VL Images + VL Names` and `Pool Images + Pool Names` is -3.9% for VL-Baseline and $+4.1\%$ for VL-Norm.

A.4 Appendix Six

Results Referential Ambiguity

Table 3: Performance of the models across different referent counts

Model Type	Referent Count	Mean Accuracy	SD	SE
VL-Baseline	2	0.811	0.063	0.014
VL-Baseline	3	0.644	0.109	0.024
VL-Baseline	4	0.600	0.132	0.030
VL-Norm	2	0.928	0.070	0.016
VL-Norm	3	0.756	0.107	0.024
VL-Norm	4	0.683	0.104	0.023
SGCL-Norm	2	0.889	0.079	0.018
SGCL-Norm	3	0.667	0.135	0.030
SGCL-Norm	4	0.606	0.080	0.018

Table 4: Summary of statistical analyses - Referential Ambiguity

Analysis	Effect	F-value	η_p^2
3×3 ANOVA (Model × Referent Count)			
	Model	$F(2, 171) = 16.457^{***}$	.161
	Referent Count	$F(2, 171) = 97.787^{***}$	.534
	Model × Referent Count	$F(4, 171) = 0.802$	.018

Note: $^{***}p < .001$. η_p^2 = partial eta squared.
Assumption checks: Shapiro-Wilk W = 0.984, p = .043; Levene's F = 2.226, p = .028.

A.5 Appendix Seven

Results Zipfian Distribution

Table 5: Performance of the models across the different frequency bins

Model Type	Frequency Group	Mean Accuracy	SD	SE
VL-Baseline	Low	0.217	0.068	0.015
VL-Baseline	Matched	0.508	0.150	0.034
VL-Baseline	High	0.600	0.249	0.056
VL-Norm	Low	0.346	0.112	0.025
VL-Norm	Matched	0.604	0.145	0.032
VL-Norm	High	0.525	0.255	0.057
SGCL-Norm	Low	0.321	0.125	0.028
SGCL-Norm	Matched	0.604	0.143	0.032
SGCL-Norm	High	0.500	0.215	0.048

Table 6: Performance of the models on matched frequency pairs across Zipfian and Uniform distribution

Model Type	Distribution	Mean Accuracy	SD	SE
VL-Baseline	Uniform	0.475	0.075	0.017
VL-Baseline	Zipfian	0.508	0.150	0.034
VL-Norm	Uniform	0.684	0.091	0.020
VL-Norm	Zipfian	0.604	0.145	0.032
SGCL-Norm	Uniform	0.591	0.067	0.015
SGCL-Norm	Zipfian	0.604	0.143	0.032

Table 7: Summary of statistical analyses - Zipfian

Analysis	Effect	F-value	η_p^2
Zipfian Frequency Gradient (3×3 ANOVA)			
	Model	$F(2, 171) = 1.297$	.015
	Frequency	$F(2, 171) = 46.412^{***}$	.352
	Model × Frequency	$F(4, 171) = 2.840^{*}$	.062
Zipfian vs. Uniform (2×3 ANOVA)			
	Model	$F(2, 114) = 17.758^{***}$	.238
	Distribution	$F(1, 114) = 0.270$	.002
	Model × Distribution	$F(2, 114) = 2.647^{\dagger}$	.044

Note: $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$, $^{\dagger}p < .10$. η_p^2 = partial eta squared.